

Assignment: Linear Programming – Mixing a Drink

A bartender wants to create a new fruit cocktail **using two ingredients**. The goal is to determine how many deciliters of each ingredient should be used so that the drink satisfies all requirements while **maximizing taste**.

Ingredients

Ingredient A (rum mixture): 40% alcohol, 5 g sugar per 1 dl, taste score 6 points per 1 dl.

Ingredient B (fruit juice): 0% alcohol, 12 g sugar per 1 dl, taste score 9 points per 1 dl.

The cocktail must satisfy the following conditions:

The total volume must be exactly 10 dl.

The drink must contain at least 20% alcohol of the total volume.

The total sugar content must not exceed 90 g.

Fruit juice must represent at least 40% of the total drink volume.



Task

Formulate the linear programming model and determine the optimal values of x and y .

Moreover, interpret your results, find vector of basic solutions, what is optimal solution, which variables are basic and non-basic in optimal solution. Calculate sensitivity analysis for right-hand side vector and cost-coefficients. Is there any alternative solution? How to interpret basic slack variables? Is it possible to solve this task graphically as well?

Decision Variables

x ... amount of ingredient A (in dl)

y ... amount of ingredient B (in dl)

Objective Function

$$\max Z = 6x + 9y$$

Constraints

1) $x + y = 10$

2) $0.40x \geq 0.20(x+y)$

3) $5x + 12y \leq 90$

4) $y \geq 0.40(x+y)$

$$x \geq 0, y \geq 0$$



TASK for extra 1 point for the exam

Only for the first five quick and correct answer. **Deadline: 15th February 2026.**

Task: Solve an example from the first lecture (in excel, all steps of Simplex algorithm). Calculate intervals of stability for vector RHS: b_1, b_2, b_3, b_4 ; calculate interval stability for cost coefficients – cx_1, cx_2 . Answer all questions from page 1. Send me an excel file by e-mail to horakovat@pef.czu.cz.